

Juan Pablo Garcia Chavez

Junior Data Scientist | Python | Machine Learning | SQL | Predictive Modeling

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Portfolio: <https://JuanPa7799.github.io/juanpablo-data-portfolio/>

Profile

Junior Data Scientist with a background in Mechatronics Engineering and a Master's degree in Electronic Engineering. I build end-to-end projects with Python, pandas, scikit-learn, exploratory analysis, feature engineering, model validation, and clear result communication.

Skills

- Python, pandas, NumPy, scikit-learn, SQL.
- Classification, regression, time series, NLP, and computer vision.
- Evaluation with ROC-AUC, F1, RMSE, MAE, and sMAPE.
- Data preparation, model comparison, and metric interpretation.

Selected Projects

Interconnect churn prediction | Applied academic project

- Context: estimated customer cancellation risk to prioritize retention in a telecom business case.
- Technologies: Python, pandas, scikit-learn, feature engineering, and GradientBoostingClassifier.
- Measurable result: 7043 rows integrated from **contract**, **personal**, **internet**, and **phone**; 0.8963 test ROC-AUC and 0.8538 test accuracy.
- Impact: built an end-to-end workflow from source integration to final model selection.

Bank churn with imbalanced classes | Applied academic project

- Context: detected bank customers at risk of churn in an imbalanced classification problem.
- Technologies: Python, pandas, scikit-learn, class balancing, and Random Forest.
- Measurable result: 0.6365 test F1, 0.865 test ROC-AUC, and 0.6904 churn recall.
- Impact: aligned the metric choice with the cost of missing at-risk customers.

Gold recovery | Zyfra | Applied academic project

- Context: predicted gold recovery in an industrial process while avoiding data leakage.
- Technologies: Python, pandas, scikit-learn, variable validation, and regression models.
- Measurable result: formula validation MAE 0.0000 and 8.49% total sMAPE.
- Impact: applied production-aware feature selection to a process-specific business metric.

Taxi order forecasting | Applied academic project

- Context: forecasted hourly taxi demand to support peak-hour operations.
- Technologies: Python, pandas, scikit-learn, temporal features, lags, and Random Forest.
- Measurable result: 43.21 test RMSE, meeting the operational target of RMSE $\leq$ 48.
- Impact: converted a time series into a supervised learning pipeline with chronological validation.

Education

- Master's degree in Electronic Engineering.
- Mechatronics Engineering.
- Applied training in Data Science and Machine Learning.

Links

- Data Scientist dashboard: <https://JuanPa7799.github.io/juanpablo-data-portfolio-/dashboards/data-scientist/>
- Main portfolio: <https://JuanPa7799.github.io/juanpablo-data-portfolio-/>